

Task - 2 : Medium

Matrix Operation

A) Matrix Multiplication

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int A[3][3], B[3][3], C[3][3], i, j, k;
```

```
    printf("Enter 3x3 elements of array A:\n");
```

```
    for(i = 0; i < 3; i++)
```

```
    {
```

```
        for(j = 0; j < 3; j++)
```

```
        {
```

```
            scanf("%d", &A[i][j]);
```

```
        }
```

```
    }
```

```
    printf("Enter 3x3 elements of array B:\n");
```

```
for(i = 0; i < 3; i++)
{
    for(j = 0; j < 3; j++)
    {
        scanf("%d", &B[i][j]);
    }
}

// Matrix multiplication logic
for(i = 0; i < 3; i++)
{
    for(j = 0; j < 3; j++)
    {
        C[i][j] = 0;

        for(k = 0; k < 3; k++)
        {
            C[i][j] = C[i][j] + A[i][k] * B[k][j];
        }
    }
}

printf("Elements in array C:\n");
```

```
for(i = 0; i < 3; i++)  
{  
    for(j = 0; j < 3; j++)  
    {  
        printf("%d\t", C[i][j]);  
    }  
    printf("\n");  
}  
return 0;  
}
```

B) Addition of Two Matrix

```
#include <stdio.h>
int main()
{
    int A[3][3], B[3][3], C[3][3], i, j, k;
    printf("Enter 3x3 elements of array A:\n");
    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            scanf("%d", &A[i][j]);
        }
    }
    printf("Enter 3x3 elements of array B:\n");
    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            scanf("%d", &B[i][j]);
        }
    }
}
```

```
}  
// Matrix addition logic  
for(i = 0; i < 3; i++)  
{  
    for(j = 0; j < 3; j++)  
    {  
        C[i][j] = A[i][j] + B[i][j];  
    }  
}  
printf("Elements in array C:\n");  
for(i = 0; i < 3; i++)  
{  
    for(j = 0; j < 3; j++)  
    {  
        printf("%d\t", C[i][j]);  
    }  
    printf("\n");  
}  
return 0;  
}
```

C) Transpose Of Matrix

```
#include <stdio.h>
int main()
{
    int A[3][3], i, j;
    printf("Enter 3x3 elements of A:\n");
    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            scanf("%d", &A[i][j]);
        }
    }
    printf("Original matrix A is:\n");
    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            printf("%d\t", A[i][j]);
        }
    }
}
```

```
    printf("\n");  
}  
printf("Transpose matrix A is:\n");  
for(i = 0; i < 3; i++)  
{  
    for(j = 0; j < 3; j++)  
    {  
        printf("%d\t", A[j][i]);  
    }  
    printf("\n");  
}  
return 0;  
}
```